EFFTRONICS SYSTEMS PVT. LTD.

Placement Drive Analysis

CSE / IT - Software Engineer Track

Final Technical Interview preparation guide for shortlisted students

1. About the company

Efftronics Systems Pvt. Ltd. (founded 1985, ~900+ employees) is headquartered in Vijayawada, with its R&D campus at Mangalagiri, Andhra Pradesh. It builds smart-city, smart-building, railway signalling and IoT products - so it hires both **software engineers** and **firmware / embedded engineers**. Freshers usually join as R&D interns or trainees.

Your role: Software Engineer / Software Developer (R&D). You will build the software side of Efftronics' IoT and smart-city products - databases, applications and logic in C/C++/Java. The interview checks fundamentals and clean hand-written code far more than advanced DSA.

2. Selection process

Stage	Format	What is tested
1. Written / online test	Online or pen-and-paper	Aptitude & maths, C programming, core subject (DBMS for CSE; DE / MPMC / Analog for ECE)
2. Technical Interview 1	Virtual or on campus (sometimes run by a third party)	Core fundamentals, resume, a small coding question
3. Office day - written papers	Pen-and-paper at Mangalagiri / Vijayawada	Psychometric test, C program writing, DBMS / SQL or Digital design & embedded C, maths
4. Medical check	In-house	Eyesight, colour blindness, height, weight
5. Final Technical Interview(s)	Face to face, often 2 panels + senior engineers	Questions **based on what you wrote in the papers** + the six subjects above
6. HR interview	Face to face	Background, relocation, agreement acceptance, attitude

Confirmed by the college for the Final Technical Interview: Digital Electronics, DBMS, Embedded C, C Programming, Algorithms and Maths. Expect all six to be touched - the depth on each depends on your track (see the subject priority table in Part B or Part C).

3. Subject priority for this track

Subject	Depth	Why
C Programming	DEEP	Longest written paper (about 1.5 hrs); every report mentions C coding
DBMS & SQL	DEEP	Separate written paper (about 40 min) + a dedicated interview round
Algorithms & Data Structures	DEEP	Sorting, searching, linked lists, complexity of the code you write
Embedded C	BASIC	Bit manipulation, volatile/static, endianness, interrupts

Subject	Depth	Why
Digital Electronics	BASIC	Number systems, gates, K-map, combinational vs sequential, flip-flops
Microprocessors	BASIC	Architecture basics, MP vs MC, RISC vs CISC
CS Fundamentals	BASIC	OOP, OS and networking basics are asked in software interviews
Maths / Aptitude	BASIC	Number theory and quantitative basics

4. Training coverage - areas to prepare

Each subject is tagged with the depth expected for this track: **DEEP** = core of the interview, expect detailed and follow-up questions; **BASIC** = working knowledge, definitions and simple problems.

C Programming DEEP

- **Basics:** data types, operators, precedence, type casting, sizeof.
- **Control flow and loops:** pattern and number problems written by hand.
- **Number logic without strings:** digit extraction, reverse, palindrome, Armstrong, prime, GCD/LCM, digit replacement.
- **Arrays:** reverse, rotate, second largest, duplicates, frequency count, merge two sorted arrays.
- **Strings (char arrays):** length, reverse, compare, copy without library functions; anagram; word count.
- **Matrices:** addition, multiplication, transpose, diagonal sums.
- **Functions and recursion:** call by value vs reference; recursion vs iteration.
- **Pointers:** arithmetic, pointer to pointer, pointers with arrays and strings, function pointers.
- **Dynamic memory:** malloc / calloc / realloc / free; memory leak; dangling pointer.
- **Structures and unions:** difference, structure padding, bit-fields.
- **Storage classes:** auto, static, extern, register. **Preprocessor:** macros vs functions, conditional compilation.
- **Output prediction:** pre/post increment, pointer tricks, printf return value.

DBMS & SQL DEEP

- **Concepts:** DBMS vs file system, data models, ER diagrams.
- **Keys:** primary, foreign, candidate, super, composite, unique.
- **SQL commands:** DDL, DML, DCL, TCL; DELETE vs TRUNCATE vs DROP.
- **Query writing on paper:** WHERE, ORDER BY, GROUP BY + HAVING, joins (inner, outer, self), sub-queries.
- **Classic queries:** Nth highest salary, employee earning more than manager, department-wise aggregates, duplicates.
- **Normalisation:** 1NF to BCNF on a sample table.
- **Transactions:** ACID with an example, COMMIT / ROLLBACK.
- **Also:** indexing, views, constraints.

Algorithms & Data Structures DEEP

- **Complexity:** Big-O (time and space) of any code you write.
- **Searching:** linear and binary search.
- **Sorting:** bubble, selection, insertion - write by hand; merge and quick sort - explain.
- **Data structures in C:** stack, queue, singly linked list (insert, delete, reverse).
- **Recursion problems:** factorial, Fibonacci, power, tower of Hanoi.
- **Theory:** basics of trees and graphs.

CS Fundamentals BASIC

- **OOP:** class, object, inheritance, polymorphism, encapsulation, abstraction.
- **Operating systems:** process vs thread, scheduling, deadlock, memory management.
- **Computer networks:** OSI model, TCP vs UDP.

Embedded C BASIC

- **Bitwise operations:** set / clear / toggle / check a bit; count set bits; power of 2; swap without temp; extract a bit-field.
- **Keywords:** volatile, const, static, extern - and why embedded code needs them.
- **Endianness:** concept and a C program to detect it.
- **Interrupts:** what an ISR is; polling vs interrupt.

Digital Electronics BASIC

- **Number systems:** conversions, 1's and 2's complement, BCD, Gray, Excess-3.
- **Boolean algebra:** De Morgan, SOP / POS; logic gates; universal gates (any gate from NAND / NOR).
- **K-map simplification** up to 4 variables.
- **Combinational circuits:** half / full adder and subtractor, mux / demux, encoder / decoder, comparator.
- **Sequential circuits:** SR, JK, D, T flip-flops; race-around condition; master-slave.

Microprocessors & Microcontrollers BASIC

- 8085 / 8086 architecture, registers, flags, addressing modes.
- Program counter, stack pointer, instruction cycle, instruction length.
- Microprocessor vs microcontroller; RISC vs CISC; Harvard vs Von Neumann.

Maths / Aptitude BASIC

- **Number theory:** divisibility, primes, HCF / LCM, remainders.
- **Arithmetic:** percentages, profit and loss, ratio, averages.
- **Word problems:** time and work, speed and distance.
- **Counting:** probability, permutations and combinations basics.

5. Written test practice

- Write C programs **by hand** with neat indentation and a dry run of sample input in the margin.
- Write SQL queries on paper - including joins and GROUP BY - and check them against a sample table.
- Take **timed mock papers** that mirror the office day: C coding (about 1.5 hrs) + DBMS / SQL (about 40 min) + maths.
- After each mock, explain your own answers aloud as the interviewer would ask.

6. Sample interview questions (from past drives + expected)

Area	Sample interview questions
C	• Replace every digit 9 in an integer with 81 without converting it to a string (1298 -> 12818). • Write matrix multiplication. • Difference between malloc and calloc?
DBMS	• Primary key vs foreign key. • Explain ACID with an example. • Find the 2nd highest salary. • DELETE vs TRUNCATE vs DROP.
Algorithms	• Write bubble sort and give its complexity. • Why does binary search need a sorted array?
Embedded C	• Write a macro to toggle the nth bit. • Why is a variable declared volatile?
DE	• Convert $(45)_{10}$ to binary and hex. • Sequential vs combinational circuit.
Maths	• HCF/LCM word problem; percentage / profit-loss question.

7. Solved example - the reported 'replace 9 with 81' question

```
#include <stdio.h>

/* Replace every digit 9 with 81 without using strings.
   1298 -> 12818                                          */
long replaceNine(long n) {
    long result = 0, place = 1;
    if (n == 0) return 0;
    while (n > 0) {
        int d = n % 10;              /* take the last digit      */
        if (d == 9) {
            result += 81 * place;    /* insert two digits '81'    */
            place  *= 100;
        } else {
            result += d * place;
            place  *= 10;
        }
        n /= 10;                     /* drop the last digit       */
    }
    return result;
}

int main(void) {
    printf("%ld\n", replaceNine(1298));   /* prints 12818 */
    return 0;
}
```

Dry run: digits are read right to left (8, 9, 2, 1). A 9 adds 81 and shifts the place value by 100 instead of 10. Be ready to explain this in the interview.

8. Interview skills and strategy

- **Defend your paper.** Interviewers open your answer sheet and ask why you wrote each line. Never write code you cannot explain - a simple, correct solution beats a clever one you can't defend.
- **Practise on paper, not an IDE.** Write neat, indented C by hand; dry-run with a sample input in the margin.
- **Say 'I don't know' honestly.** Past selected candidates admitted not knowing 1-2 answers and were still selected. Bluffing is caught quickly. Add: 'but this is how I would approach it...'
- **Think aloud.** For design or coding questions, state the approach, edge cases and complexity before writing.

- **Long day - plan for it.** The office day can run from about 8:30 AM to 6 PM. Carry ID, resume copies, pens, a water bottle; stay sharp through the afternoon panels.
- **Know your resume project cold** - block diagram, your exact role, one problem you solved.
- **Self-introduction:** prepare a crisp 1-minute version - name, college, branch, key skills, one project, why Efftronics.
- **Psychometric test:** answer honestly and consistently; the same trait is asked in different ways, so contradictions show.
- **HR round:** be ready for relocation to Andhra Pradesh, the service agreement, 'why Efftronics' and your long-term plan.
- **Mock interviews:** take at least one full mock technical interview with feedback before the drive.

9. Drive briefing - offer, agreement and logistics

- Freshers typically join as **R&D Intern / Trainee Engineer**, then move to engineer after the training period.
- Published offer letters (2020-2022) show packages from about **Rs. 3.98 LPA** (firmware trainee, 2020) to a gross of **Rs. 35,888 - 47,088 per month** (software R&D interns, 2022). Current figures may differ - confirm with the placement cell.
- **Service agreement:** offer letters mention a 2-year service period during which resignation is not allowed; candidate reports also mention 2.5-3 year agreements, retention of original certificates, and part of the salary being held back until completion.
- The offer letter has to be signed by you **and your parents**. Read the terms fully and decide *before* the final day - backing out after an offer hurts your college's future drives.
- **Office-day logistics:** a long day (about 8:30 AM - 6 PM) with written papers, a medical check (eyesight, colour vision, height, weight) and interviews.
- **Carry:** ID proof, college ID, several resume copies, pens, and certificates if asked by the placement cell.

10. Checklist

	Final-week checklist
[ ]	Write 15+ programs by hand and dry-run each one
[ ]	Revise every subject in the priority table at least once
[ ]	Prepare a 1-minute self-introduction and a 2-minute project explanation
[ ]	Practise one mock 'defend your paper' interview with a friend or trainer
[ ]	Read and discuss the service agreement with your parents
[ ]	Keep ID proof, resume copies, college ID, pens and certificates ready

Note on sources: Compiled from candidate interview reports on Glassdoor and Naukri Campus (2020-2026), published Efftronics offer letters, the company profile, and the college's notice to shortlisted students. Patterns vary by college and year - treat this as a preparation guide, not an official syllabus.